

- Q.5 An auto-transformer is different from a two-winding transformer because-
- It has no core
 - It has only one winding
 - It uses AC only
 - It is used in DC circuits
- Q.6 Which power supply component used to regulate power spikes
- Stabilizer
 - Transformer
 - UPS
 - Battery
- Q.7 Which device store energy for backup in UPS
- Generator
 - Inverter
 - Capacitor
 - Battery
- Q.8 Star-delta connections utilized for mainly
- motors only
 - transformers
 - signals processing
 - lighting circuits
- Q.9 Earthing is required for
- signal transmission
 - voltage control
 - Patient safety and fault protection
 - None
- Q.10 Which connectors used in printers
- RJ-45
 - BNC
 - FRC
 - DIN

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Insulation.
- Q.12 Transducer
- Q.13 Micro-shock
- Q.14 Expand CMC
- Q.15 Expand EEG
- Q.16 Name one non-invasive biomedical sensor.
- Q.17 Leakage current
- Q.18 Expand CVT
- Q.19 Line regulation
- Q.20 Calibration

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain main precautions required for installing medical equipment in hospitals.
- Q.22 Explain Grounding and explain its significance.
- Q.23 Write a note on importance of proper spacing when handling medical equipment.
- Q.24 Write a note on coaxial cable and twisted pair cable.
- Q.25 Explain types of electrical cables used in biomedical systems.
- Q.26 Explain BNC connector & its importance.
- Q.27 Write a note on RJ-45 connector & its significance.